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MCA Major Project

(2023 – 2025 Admitted Batch)

Report Submission for Zeroth Review

(Abstract, Literature Survey, and Problem Statement)

Deep Learning Approach For Accident Hotspot Identification Using GeoCrashNet

Submitted By

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Title of the Project: Deep Learning Approach For Accident Hotspot Identification Using GeoCrashNet

Abstract:

Road accidents are a significant concern worldwide, leading to loss of lives, injuries, and economic setbacks. Identifying accident hotspots accurately is crucial for implementing effective road safety measures. GeoCrashNet is a deep learning-based framework for accident hotspot identification. The model leverages historical accident data, geospatial features, and environmental factors to predict high-risk locations.

GeoCrashNet integrates Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to process spatial and temporal accident data. CNNs capture the geospatial distribution of accidents, while RNNs analyze temporal accident trends. Additionally, the model incorporates Graph Neural Networks (GNNs) to represent road networks and connectivity patterns effectively. The dataset consists of historical accident reports, road topology, weather conditions, and traffic flow data. Using GPS coordinates and clustering techniques, the model identifies high-risk zones and ranks them based on accident severity and frequency.

Experimental results demonstrate that GeoCrashNet outperforms traditional statistical models and machine learning approaches in accident hotspot detection. The proposed system can assist urban planners, traffic authorities, and policymakers in designing safer road infrastructures and implementing proactive accident prevention measures. By leveraging deep learning and geospatial data, this research aims to enhance road safety and reduce accident occurrences. Future improvements may include real-time accident forecasting and integration with smart city traffic management systems.

Keywords

Deep Learning, Accident Hotspot Identification, GeoCrashNet, Geospatial Analysis, CNNs and RNNs, Graph Neural Networks (GNNs), Traffic Prediction, Spatial-Temporal Analysis.

Literature Survey:

S. No	Author(s)	Year	Methodology	Findings	Limitations
1	Gutierrez-Osorio & Pedraza	2020	Review of deep learning methods	High accuracy in accident prediction	No empirical results
2	Szenasi & Csiba	2020	DBSCAN clustering on GPS data	Identified high-density accident zones	Inconsistent results
3	Zarei et al.	2021	CGAN-EB model	Performed well in low-sample scenarios	Needs real-world validation
4	Zhang et al.	2021	TCFGC-Net with satellite and street view data	93.7% accuracy in hotspot detection	Limited to urban areas
5	Alotaibi et al.	2021	Video-based accident detection	Effective for visual accident recognition	Cannot predict accidents
6	Al-Bdour et al.	2022	GIS-based spatial analysis	Identified accident clusters in urban zones	Region-specific findings
7	Oluwasegun et al	2022	Kernel density estimation in GIS	Located highway accident hotspots	Limited to a specific highway
8	Al-Ghamdi	2022	Spatial autocorrelation analysis	Accidents frequent in city centers	Region-specific results
9	Li et al	2023	SFRAN for street view imagery	Effective in dynamic accident hotspot detection	Requires high-quality imagery
10	Kumar & Toshniwal	2023	Data mining for accident analysis	Defined accident-prone locations	Review study only
11	Montella	2023	Comparison of HSID methods	Empirical Bayes method was most reliable	No new methodology
12	Theofilatos et al	2023	Traffic flow and speed analysis	Identified key crash factors	Focuses only on PTW crashes
13	Erdogan et al	2022	GIS for accident hotspots	Identified high-density areas	Review study only

14	Deepthi & Ganeshkumar	2023	GIS-based hotspot detection	Effective spatial accident analysis	Region-specific findings
15	Roshandel et al.	2024	Classification models for prediction	Deep learning sometimes outperforms simpler models	No empirical data

Problem Statement:

Problem Definition:

The objective of this project is to develop a deep learning model, GeoCrashNet, to analyze large-scale accident data and accurately detect high-risk accident-prone locations. The model will incorporate geospatial features, road network attributes, real-time traffic conditions, and weather data to improve accident hotspot detection.

Importance of the Problem & Its Impact:

Accurate accident hotspot detection enhances road safety through proactive measures and better traffic management. AI-driven insights help allocate resources efficiently, reducing accidents and economic losses. Real-time monitoring improves risk assessment and emergency response. GeoCrashNet ensures scalability, making transportation safer and smarter.

Challenges in Solving the Problem:

1. **Data Availability and Quality** – Accident records, traffic data, and weather reports must be accurate and complete for effective model training. Inconsistencies and missing data can impact model performance.
2. **Geospatial Data Complexity** – Integrating various spatial features (e.g., road structures, intersection types, pedestrian density) requires sophisticated data preprocessing techniques.
3. **Real-Time Data Processing** – Handling live traffic and environmental data streams in real-time poses computational and algorithmic challenges.
4. **Interpretability of Deep Learning Models** – Unlike traditional statistical methods, deep learning models function as black boxes, making it difficult to explain why certain locations are identified as hotspots.
5. **Generalization Across Different Regions** – Accident patterns vary across regions due to infrastructure, traffic laws, and driving behavior, requiring model retraining and fine-tuning for different locations.

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